

V4 response working draft

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INTERNAL PENDING WORK

This response is a Stage 2 working revision for submission 0170c7b2-1065-4344-a5d5-b839b51ac22c. The manuscript title is *RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text*. The original review letter is preserved below. Genuine pending items remain marked. In particular, semantic validation and final V4 archive coverage are unresolved.

Editor letter

— Dear Dr Mantzaris,

Your manuscript, *"RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text"*, has now been assessed.

We invite you to revise your paper, carefully addressing the comments from the reviewers and the editor. Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained. When your revision is ready, please submit the updated manuscript and a point-by-point response. This will help us move to a swift decision.

Please note that if your manuscript uses any custom or bespoke computational tool or code, or reports a new algorithm, tool, software, or a pipeline (even if individual components are not new), the underlying code must be deposited in a recognised DOI-assigning repository (e.g. zenodo) and linked either from Methods or a dedicated Code Availability section.

Editor Comments

"The manuscript presents a technically interesting and systematically evaluated framework for embedding cryptographic payloads into language-model-generated text through deterministic token-rank manipulation. Reviewer 2 considers the previous concerns largely addressed. However, several substantive issues remain before the conclusions can be fully supported. In particular, the authors should clarify the robustness boundaries under text transformations and configuration desynchronization, analyze the implications of zero cross-model recovery, explain the near-perfect detectability under an exact-model-aware attacker, strengthen semantic-coherence validation, document all deterministic filtering rules, and provide deeper analyses of entropy-gating failures, computational complexity, and quantization-induced divergence. Suggested references are optional, and the authors may select appropriate recent peer-reviewed literature." -Chun-Wei Yang

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Purva Dhamankar —

editor _accuracy response, complete.

Direct answer. We have revised the claims to distinguish surface-form concealment, exact configured recovery, visible-text transport, detectability and reader-perceived naturalness. Adverse and inconclusive outcomes remain visible.

Evidence and change. The V4 text corrects the historical raw and limited unmodified references to saved-token recovery, documents the unfiltered sampled-skip entropy experiment, and separates the directional fixed-message quantization endpoint from free generation. The final numerical check links statements to retained tables.

Locations. Abstract, main Results under Boundary validation and targeted diagnostics, Discussion, and Supplementary Notes S16 through S19.

archive response, pending.

Direct answer. The underlying code has a published DOI. Version 2.0.0 is 10.5281/zenodo.22555497 and covers V3, not the V4 work. We retain that correct published citation. A new V4 deposit has not been published.

Evidence and change. The archive audit verified the published contents and found that its description incorrectly excludes manuscript files. We prepared a local inventory and a corrected description for a future version, including the V4 source, evidence and document files.

Locations. Main Data and Code availability. Local release specification and prospective inventory.

INTERNAL PENDING WORK. Assemble from the exact finalized commit, verify the archive inventory, publish an authorized new version, and insert its real DOI only after coverage is checked. No deposit or submission was made in this stage.

editor _technical response, partial.

Direct answer. The revision adds targeted transport and quantization endpoints, the full filter specification, all six entropy-failure diagnostics, a configuration compatibility check and source-based theoretical analysis. Semantic validation remains incomplete.

Evidence and change. Both blinded pilots support conditional context gain but do not validate the semantic-relatedness metric. We retain that limitation and label the subsequent likelihood study exploratory. We do not claim that automated scores prove reader judgments.

Locations. Main Methods under Boundary instrumentation and prospective control validation, targeted Results, and Supplementary Notes S16 through S19.

INTERNAL PENDING WORK. R1.4 remains partly unresolved because the prospective two-metric control-validation requirement was not met. Final archive coverage is also pending.

Reviewer 2 acknowledgment

Reviewer 2

Thank the authors for addressing most of my concerns.

reviewer2 response, complete.

Direct answer. We thank Reviewer 2 for acknowledging the earlier revisions.

Evidence and change. The acknowledgment is retained in full with the current reviewer label. This revision responds separately to the new editor and Reviewer 1 requests.

Locations. This response, Reviewer 2 acknowledgment. No manuscript change is needed for this acknowledgment.

Reviewer 1 general assessment and wording instruction

Reviewer 1

The manuscript proposes a steganographic method embedding cryptographic payloads into language model outputs via deterministic token rank manipulation. While demonstrating exact recovery under strictly controlled conditions, the framework exhibits fragility against text transformations and cross-model decoding, limiting its practical utility despite comprehensive evaluations of capacity and neural steganalysis detection.

Overall, the study presents a reproducible measurement framework for linguistic steganography. However, its reliance on perfectly synchronized configurations and failure during standard markdown operations require deeper theoretical justification. Furthermore, near-perfect detectability by exact-model-aware attackers and the absence of perceptual validation weaken its broader claims. Significant revisions addressing robustness boundaries, theoretical limitations of detectability, and expanded discussions on quantization sensitivity are required before publication can be considered.

Please do not arbitrarily simplify or rephrase the review comments, as this hinders the assessment of your revisions.

general response, complete.

Direct answer. We agree that recovery requires a synchronized configuration and that the observed transport and detection failures substantially restrict practical claims. The revision presents these as limitations, not as problems solved by a new robustness method.

Evidence and change. The discussion states that a fully configured attacker may decode, distinguishes trace-informed detection from visible-text scoring, and labels the block/ECC fallback as unimplemented. The unsuccessful semantic pilots remain visible. The additional transport diagnostics are illustrative rather than population estimates.

Locations. Abstract, main Discussion under Configuration sensitivity, detection and a proposed fallback, limitations, and Supplementary Notes S17 through S19.

wording response, complete.

Direct answer. We have retained the complete review wording without simplification or paraphrase, including both general paragraphs and this instruction.

Evidence and change. The thirteen exact quote blocks reconstruct the original request file. The validator checks each escaped block against the unchanged source and checks the original file hash. Answers are separate from quoted comments.

Locations. All review-comment blocks in this response. The original requests file and exact-block ledger remain unchanged.

Reviewer 1 comment 1

1- In the abstract and introduction, the manuscript admits severe transmission fragility. While you explicitly scope this as a measurement framework, you must expand the theoretical discussion on why standard operations like markdown copy cause complete failure and discuss potential boundary mitigations.

R1.1 response, complete.

Direct answer. The tested Markdown operation is a specific composite transform. It adds a blockquote prefix to each line and removes whitespace. The receiver then applies saved token offsets after retokenization. This can shift the forced span, change tokens and alter contextual ranks. It does not demonstrate failure for every Markdown workflow.

Evidence and change. Five hash-selected sources give 25 decomposed arms and 135 segment instances. Exact reuse leaves 26 missing segment replays. We also checked all 53 reused segment inputs at the new context allocation, with identical rank and error outcomes. These checks add no logical cover arms. Prefix-only insertion recovers none of the five. Declared wrapper removal recovers one originally failing Mistral case, but cannot restore removed whitespace. Byte edits, extracted spans, first rank differences, errors and recovered bytes are retained. Original composite recovery remains 0/144.

Locations. Main Methods under Decoder compatibility and bounded transport diagnostics, targeted Results, and Supplementary Note S18 with its illustrative recovery table.

Reviewer 1 comment 2

2- Within the bounded payload methods section, assuming perfectly synchronized model configurations is highly restrictive. Please introduce a theoretical discussion addressing how minor configuration desynchronization, such as slight tokenizer version updates, theoretically impacts the overall payload recovery rate.

R1.2 response, complete.

Direct answer. A minor version change has no universal recovery-rate implication. It may preserve the relevant tokenization or change token identities, offsets and conditioning. Model or gate differences introduce other disagreement mechanisms. We state the required shared contract and conditional agreement argument explicitly.

Evidence and change. A new canonical configuration fingerprint includes eleven required contract components. Tests accept matching configurations, verify deterministic serialization and reject changes to every component. It is a compatibility API, not authentication or a robustness guarantee, and was not retroactively applied to the historical messages.

Locations. Main Methods under Decoder compatibility and bounded transport diagnostics, main Discussion under Configuration sensitivity, detection and a proposed fallback, and Supplementary Note S19.

Reviewer 1 comment 3

3- In the exact copy fragility evaluation, the zero percent recovery rate for cross-model decoding is a severe limitation. You should propose and discuss a conceptual error correction strategy or fallback mechanism to mitigate this lack of robustness during mismatched decoding.

R1.3 response, complete.

Direct answer. Zero cross-model recovery remains a severe limitation. We propose configuration identification before decoding, independently framed blocks with context resets, integrity checks, bounded outer error/erasure correction and retransmission. This is a conceptual fallback, not an implemented improvement.

Evidence and change. The argument states the minimum-distance condition $2e+s<d$ and its assumptions. ECC cannot reconstruct an arbitrary wrong model codebook or repair persistent synchronization loss. Reliable framing, a supported configuration and overhead for redundancy and retries remain necessary. Retransmission under an unresolved mismatch is ineffective.

Locations. Main Discussion under Configuration sensitivity, detection and a proposed fallback, and Supplementary Note S19.

Reviewer 1 comment 4

4- Throughout the segmented multi-cover method section, the distinction between payload-bearing spans and natural tails lacks perceptual validation. You must provide stronger automated semantic coherence metrics to prove that these generated transitions appear natural and logically connected to readers.

R1.4 response, partial.

Direct answer. We tested complementary semantic-relatedness and independent-model conditional-compatibility metrics. The semantic metric did not pass the prospective control-separation requirement in either of two blinded pilots. Context gain did pass. We therefore cannot claim that the requested transitions are proven natural or logically connected to readers.

Evidence and change. The pilots each cover 36 identities. Initial semantic separation was 0.0078 with interval [-0.0416, 0.0572]. The one predeclared 16-token amendment on disjoint recipient and donor payloads gave -0.0106 [-0.0728, 0.0519]. Both context-gain intervals were positive. The frozen exploratory study completed 2,750 boundaries across 92 payloads. Actual minus ordinary context gain was 0.1575 nats per tail token, with a 95% interval [0.1098, 0.2066]. Actual minus shuffled gain was 2.3265 [2.2705, 2.3912]. Eligibility, fixed-corpus intervals, donor reuse and influence, and predeclared weak, typical and strong examples are reported. The ordinary comparison also differs in continuation policy, sampled controls versus greedy RankCloak tails. It does not isolate a causal boundary effect. This is not a completed two-metric coherence evaluation.

Locations. Main Methods under Boundary instrumentation and prospective control validation, targeted Results and its context-gain table, and Supplementary Note S17.

INTERNAL PENDING WORK. Semantic and perceptual support remains unresolved. No further metric search was undertaken after the allowed amendment. Any future design needs a new prospective rationale or reader study, not relabeling this negative validation as success.

Reviewer 1 comment 5

5- Neural steganalysis results indicate near-perfect detection by the exact-model-aware attacker. Rather than a full redesign, you must deeply analyze the theoretical bottlenecks causing this glaring statistical detectability and clearly define the framework’s security boundaries against this specific threat model.

R1.5 response, complete.

Direct answer. The evaluated model-aware attack is trace informed and exposes a strong statistical limitation. Its features come from saved token likelihoods under the generating model. Reproducing them requires the original token path, model/backend, prompts, boundaries, resets and probability convention. A visible-text-only scorer is a distinct condition.

Evidence and change. The added discussion explains that payload-rank distributions need not match ordinary sampling. It distinguishes untempered full-vocabulary scores from the temperature-0.8 top-p-0.95 sampler and states the support conditions for a sequence-KL argument. AUC is not converted to KL, and no unique causal feature attribution is claimed. An attacker with the complete receiver configuration may decode. No confidentiality or informed-attacker indistinguishability is asserted.

Locations. Main steganalysis Methods and Discussion under Configuration sensitivity, detection and a proposed fallback. The source-based argument is in the Stage 2 theory audit.

Reviewer 1 comment 6

6- The discussion of entropy gating notes that the strict gate condition resulted in maximum-budget payload completion failures. You should investigate and report the exact linguistic contexts or token-level characteristics that triggered these specific failures during the generation loop.

R1.6 response, complete.

Direct answer. All six strict-gate failures are retained and analyzed at their original budgets. Five are Qwen and one Mistral, all ASCII B16. They consume 110/128, 116/128, 54/64, 106/128, 59/128 and 116/128 ranks. Strict completion remains 114/120.

Evidence and change. The supplement reports exact pinned-tokenizer windows, longest below-threshold runs of 50, 163, 39, 35, 267 and 51 positions, rolling progress, margins and rank pressure. The longest runs contain 30 to 157 distinct token IDs, at most three identical IDs consecutively, and rank-one frequencies of 85.7 to 98.8 percent. All 605 positions remain ordinary sampled skips with no payload progress. Twelve matched ungated/moderate trajectories and three hash-selected strict successes are retained. Observed contexts include formatted public-service and project-update passages, but low-entropy runs also occur in successes. These are descriptive associations, not established linguistic causes. The trajectories used no token filter and sampled ordinary skips, not greedy skips.

Locations. Main targeted Results and Supplementary Notes S13 and S16, including the six-case table, trajectory figure and exact local windows.

Reviewer 1 comment 7

7- The computational overhead analysis reports significant performance bottlenecks. Instead of exhaustive hardware baselines, please provide a theoretical computational complexity analysis comparing your rank transcoding method against standard linguistic steganography tools to mathematically justify the observed processing costs.

R1.7 response, complete.

Direct answer. The new comparison separates model work from sorting, rank counting, probability passes and coding operations. Rank selection sorts all allowed candidates, whereas recovery of a known token counts greater logits and lower-ID ties.

Evidence and change. The analysis inspects the primary arithmetic-coding and SAAC implementations and ADG grouping algorithm. It includes repeated diagnostic sorts, finite-precision cumulative tables, grouping search/removal, prompt prefills, KV state, logits_all storage, tails, sampled skips, segmentation and payload-side direct-subword passes. Bounds are implementation-specific and imply no cross-backend wall-time superiority.

Locations. Main Computational overhead and generalizability and Supplementary Note S19, implementation-specific operation-count table.

Reviewer 1 comment 8

8- In the reference list, citation 16 is an unverified preprint. If formally published, update the citation; otherwise, provide detailed theoretical justification for relying on its entropy-guided watermarking claims, or replace it with a peer-reviewed publication to strengthen your foundation.

R1.8 response, complete.

Direct answer. The bibliography key Cai2025EntropyGuidedWatermarking refers to the previous reference 16, arXiv:2504.12108. The inspected primary record lists no journal reference. We have replaced its role in the V4 argument with the peer-reviewed Lee and colleagues ACL 2024 paper on entropy-threshold selective watermarking.

Evidence and change. The V4 prose now cites Lee2024SWEET for the limited thresholding precedent. It distinguishes detecting a watermark from recovering a chosen payload. Neither a watermark theorem nor the unverified preprint is used as a RankCloak recovery guarantee. The historical experiment itself is still documented by its original configuration and data.

Locations. Main Introduction, Supplementary Note S13 and the V4 bibliography.

Reviewer 1 comment 9

9- The methods section describing the deterministic token filter leaves the rules for excluding markup fragments obscure. You must explicitly list these complete exclusion criteria in the main text so independent researchers can accurately replicate your experimental setup.

R1.9 response, complete.

Direct answer. The main Methods now list every literal blocked substring and every other safe_text_filter_v1 exclusion criterion from the actual unchanged code.

Evidence and change. The specification covers empty and replacement pieces, ASCII controls and exceptions, whitespace/start rules, backslashes, individual-token decoding and exceptions, mask application, special-token distinctions and stable tie-breaking. It distinguishes full selection sorting from counted rank recovery and notes limitations for concatenated markup and nonfinite logits. The versioned source link and source hash are retained in the audit, with focused edge-case tests.

Locations. Main Methods under Complete deterministic token filter specification and its complete rule table.

Reviewer 1 comment 10

10- The paired quantization experiment shows independently generated token paths diverging significantly despite identical inputs. Please expand this section to theoretically discuss how minor quantization errors in logits are amplified by the deterministic sorting process, triggering this avalanche effect.

R1.10 response, complete.

Direct answer. We separate same-history logit perturbations from subsequent feedback when independently generated token paths split. For a fixed history and allowed set, pairwise gaps above twice a uniform absolute perturbation bound preserve order. Smaller gaps and ties have no such protection. This is a conditional bound, not a measured quantization-error distribution.

Evidence and change. The audit reconciles 69,528 rank changes across 244,440 positions in 1,920 pairs. Encoded messages contribute 56,321/122,220 and ordinary controls 13,207/122,220. The distinct offline fixed-message endpoint gives zero exact Q4-to-Q8 payload recoveries in 960 supported rows, with 766 invalid-rank trials and 194 valid but wrong decoded payloads. It supports no reverse-direction claim. Observed-token replay feeds received cover tokens, not decoded payload symbols, back into the model. Existing evidence supports these claims without new neighboring-logit replay.

Locations. Main targeted Results and Supplementary Notes S14, S18 under Directional cross-quantization endpoint, and S19.